

Aditya Kulkarni

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Raleigh, NC (open to relocation)

PROFESSIONAL SUMMARY

Full stack engineer with production AI/ML experience in TypeScript and Node.js. At Isomer AI, built an AI-powered automation platform from 0 to production: multi-model LLM routing across Claude, GPT-4o, and Gemini via MCP tooling, real-time event pipelines, document processing, and workflow orchestration -- 394 commits and 7 deployments in a small team. Built a privacy-first RAG system for fintech environments with per-tenant data isolation. Fast at shipping -- 3 complete MVPs in 48-hour hackathon builds. MS CS at NC State (May 2026). Portfolio: kulkarniaditya.com

CORE COMPETENCIES

TypeScript / Node.js

React / Next.js

LLM Pipelines (Claude, GPT-4o, Gemini)

MCP Tooling

PostgreSQL / Redis

AWS

Docker / CI/CD

REST APIs

Temporal.io Workflow Orchestration

RAG / Vector Databases

Observability (Sentry, Prometheus, OTel)

Distributed Systems

EXPERIENCE

Isomer AI May 2025 - Present

Product Engineer Intern

- Shipped 10+ backend features across 7 production deployments for an AI-powered claims automation platform; 394 commits in TypeScript, Node.js, Express, PostgreSQL, Redis, and Prisma with real-time event pipelines (Pusher) and observability via Sentry
- Built multi-model LLM routing across 3 providers (Anthropic Claude, OpenAI GPT-4o, Google Gemini) and 8+ models via MCP tooling; implemented Temporal.io workflow orchestration, RBAC, document processing pipelines (PDF generation, email via Microsoft Graph API, S3 storage), and Auth0 authentication

NC State University April 2025 - Present

Graduate Research Assistant

- Designed a cloud-based genomic data platform (JBrowse 2 on Google Cloud Run) serving 500GB datasets to 100+ researchers worldwide; improved ML model accuracy by 12% via hyperparameter tuning

Mphasis (HP Company) Jun 2024 - Aug 2024

Software Developer Intern

- Built 5+ AI-driven workflows including a health claim automation tool (OCR, NLP, ML); improved claim accuracy by 25%; automated deployments on AWS and GCP with Grafana monitoring

Capgemini Jun 2023 - Aug 2023

AI/ML Intern

- Built distributed data pipelines (Hadoop + Spark batch, Kafka streaming) for support ticket automation; reduced processing time by 15% and latency by 20%

PROJECTS

Glance AI - Privacy-First RAG System Fintech / Healthcare AI

Per-tenant data isolation architecture at the vector retrieval layer for regulated environments (fintech, healthcare). Each retrieval context is scoped to its tenant boundary before LLM input -- prevents cross-tenant contamination and hallucinations. Deterministic output patterns for compliance-ready factual correctness.

Python, LLMs, Vector Databases, RAG

AlgoPulse - Agentic AI Platform Temporal + Kafka

Identified API-polling bottleneck under AI inference load; re-architected to event-driven streaming with Supabase Edge Functions. Temporal.io for durable workflow state; Kafka for fault-tolerant worker coordination. Reached 2,000 async inference requests/second with zero message loss.

TypeScript, Node.js, React, Supabase Edge Functions, Temporal.io, Kafka

PulseRank - Real-Time Leaderboard & Feed Distributed Systems

Event-driven multi-store platform: MySQL outbox + idempotency keys, MongoDB TTL projections, Redis sorted sets for sub-ms reads; live updates over Server-Sent Events. Multi-store consistency validated with Testcontainers; benchmarked with k6; OpenTelemetry + Prometheus observability.

TypeScript, NestJS, Next.js, MySQL, MongoDB, Redis, Docker, OpenTelemetry, Prometheus

LLM Eval Harness AI/ML Ops

Vendor-neutral benchmark comparing Claude Sonnet 4.6, GPT-4o, and Gemini 2.5 Pro on a 2,000-example frozen test set. Tracked hallucination rate, p50/p95 latency, and cost/call per model with statistical significance testing. DuckDB + Parquet analytics; Streamlit dashboard.

Python, Anthropic/OpenAI/Google SDKs, DuckDB, Parquet, Streamlit, Docker

Inventory Reservation Service Java / Spring Boot

Transactional inventory reservation API in Java 21 + Spring Boot preventing overselling via PostgreSQL row-level locking across multi-SKU reservations. Concurrency correctness verified with Testcontainers: two concurrent requests against quantity 1 produce exactly one success and one 409 Conflict. Flyway migrations, Spring Security RBAC, GitHub Actions CI.

Java 21, Spring Boot 3.5, PostgreSQL, Flyway, Testcontainers, Docker

PerCent - GenAI Finance App HackNC24 Top 10

Built a complete RAG-based GenAI finance app in 48 hours: goal tracking, expense logging, and an AI chatbot using Qwen-14B-Chat via Hugging Face. Ranked top 10 of 100+ teams; reduced manual financial research by 30%.

Python, Selenium, Hugging Face, Qwen-14B-Chat, RAG

EDUCATION

North Carolina State University -- MS Computer Science	Aug 2024 - May 2026
Courses: Artificial Intelligence, Operating Systems, Data Science, Software Engineering, Algorithms	
Mumbai University -- BTech Computer Science	Aug 2020 - May 2024
CGPA: 9.32/10 Machine Learning, Big Data, DS & Algorithms, Blockchain	

AWARDS & LEADERSHIP

HackNC24 Top 10 -- GenAI Finance App (PerCent); top 10 of 100+ teams, 48-hour build	2024
Sanjeevani Health App -- Top 1% of 1,100+ submissions; 87% ML accuracy	2023
IEEE Publication -- Mapify: T5 pipeline, 60% review time reduction, 500+ attendees	2022
Google Developer Student Club Lead -- 4+ events, 500+ students	2022-2024

TECHNICAL SKILLS

Languages: TypeScript, Python, Java, JavaScript, SQL, R, C **Frontend:** React, Next.js, Tailwind CSS

Backend: Node.js, Express, NestJS, FastAPI, Spring Boot, REST APIs

AI/ML: MCP, LangChain, Hugging Face, PyTorch, RAG, OpenAI/Anthropic/Google SDKs

Databases: PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch, Qdrant **Infrastructure:** AWS, GCP, Docker, Kubernetes, Kafka, Temporal.io, Terraform

Observability: Sentry, Prometheus, Grafana, OpenTelemetry, Datadog (familiar) **DevOps:** GitHub Actions, Docker Compose, Testcontainers